



THE THREE CADENCES OF PHYSICAL AI

Asynchronous Multi-Rate Temporal Hierarchy: Intent $\rightarrow$ Action Chunks $\rightarrow$ Microsecond Reflexes

SYSTEM 2: INTENT

Slow Cadence: 0.5–2 Hz (500–2000 ms)

Role: Open-vocabulary reasoning

Model: Multimodal VLM (7B–70B)

Output: Expiring Intent Lease $\mathcal{L}_{\text{intent}}$

Authority: Untrusted Proposal

Failure Mode: If VLM stalls or drops frames, lease expires harmlessly.

Lease

SYSTEM 1.5: TRAJECTORY

Medium Cadence: 20–50 Hz (20–50 ms)

Role: Action Chunk unrolling

Model: Diffusion ACT / Policy

Output: Horizon of H waypoints

Authority: Candidate Trajectory

Advantage: Amortizes neural compute delay across continuous execution.

Chunks

SYSTEM 1: REFLEX

Fast Cadence: 1000 Hz ($1.0 \text{ ms} \pm 5 \mu\text{s}$)

Role: Control Barrier Enforcement

Model: Bare-Metal FreeRTOS CBF

Output: Permitted Gate PWM (u_t)

Authority: **SOLE PERMISSION**

Safety Invariant: Deterministic emergency halt on any timing fault.